

WILMS

Women's Interest-Free Loan Management System

Notification & Communication Book — v1.7.3

WILMS Notification & Communication Book

Version: 1.7.3 Classification: Confidential

1. Overview

WILMS delivers multi-channel notifications and a Communications Center for programme-wide messaging.

2. Notification channels

| Channel | Use cases |

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| In-app | All event types; notification inbox |

| Email | Login alerts, invitations, expense reviews |

| SMS | Field alerts, payment confirmations |

3. Notification guarantees

Deduplication

Same event within deduplication window produces single notification. Prevents alert fatigue from retry or duplicate dispatch.

Quiet hours

Organisation-configured quiet hours suppress non-critical notifications. Critical security alerts (login from new device) may override.

Cron dispatch

Daily batch dispatch at 06:00 UTC via Vercel Cron (/api/cron/notifications). Failed dispatches logged; retry on next cycle.

4. Event types

| Event | Channels | Recipients |

|-----|-----|-----|

| Login alert | Email | User |

| Invitation accepted | In-app | Admin |

| Expense submitted | In-app, Email | Reviewers |

| Expense approved/rejected | In-app, Email | Submitter |

| Loan approved/rejected | In-app | Officer, Approver |

| Reconciliation variance | In-app | Admin, Approver |

| Overpayment review | In-app | Approver |

| Ops incident | In-app | Admin |

| Maintenance window | In-app | All users |

5. In-app notification inbox

Accessible from navbar bell icon. Shows unread count. Mark individual or all as read. Role-aware filtering — users see only relevant notifications.

6. Communications Center

Templates

Reusable message templates with variable substitution (e.g., {borrowerName}, {amount}). Admin creates and manages templates.

Broadcasts

Targeted messages to role groups or all users. One-way communication from HQ to field.

Message threads

Two-way messaging between HQ staff and field collectors. Threaded conversation view.

Audit

All sent messages logged with sender, recipients, timestamp, and content hash.

7. Configuration

Super Admin! Settings:

- Notification preferences per channel
- Quiet hours schedule
- Email/SMS provider credentials (environment variables)

8. API endpoints

| Method | Path | Description |

|-----|-----|-----|

| GET | /notifications | List user notifications |

| PATCH | /notifications/:id/read | Mark read |

| POST | /communications/broadcasts | Send broadcast |

9. Planned enhancements (v1.9+)

- Scheduled report delivery via notification
- Advanced routing rules
- Borrower SMS reminders (v2.5)
- Push notifications via PWA service worker

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